THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The middle term of an AP with 11 terms, first term 8 and last term 48, is: 1

(a) 28 **(b)** 24 **(c)** 32 **(d)** 48

2. If $a_{12} = 32$ and $d = 2$, then $a =$ 1

(a) 8 **(b)** 54 **(c)** 10 **(d)** 12

3. The 15th term of the AP whose 1st term is 11 and 6th term is 21 is: 1

(a) 37 **(b)** 30 **(c)** 39 **(d)** 41

4. The number of terms in the AP $2, 8, \ldots, 98$ is: 1

(a) 18 **(b)** 17 **(c)** 16 **(d)** 19

5. The sum of all two-digit multiples of 5 is: 1

(a) 850 **(b)** 1045 **(c)** 950 **(d)** 945

6. The sum of the first 12 terms of the AP $30, 27, 24, 21, \ldots$ is: 1

(a) 156 **(b)** 324 **(c)** 162 **(d)** 165

7. The middle term of an AP with 9 terms, first term 2 and last term 58, is: 1

(a) 23 **(b)** 58 **(c)** 30 **(d)** 37

8. If $a_6 = 32$ and $d = 5$, then $a =$ 1

(a) 57 **(b)** 2 **(c)** 12 **(d)** 7

9. The 10th term of the AP whose 1st term is 3 and 3rd term is 13 is: 1

(a) 43 **(b)** 53 **(c)** 50 **(d)** 48

10. The number of terms in the AP $11, 16, \ldots, 136$ is: 1

(a) 26 **(b)** 27 **(c)** 28 **(d)** 25

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum of the first 26 terms of the AP $-12, -8, -4, 0, \ldots$ 2

12. Find the sum of all natural numbers between 10 and 200 that are divisible by 5. 2

13. Which term of the AP $-3, 2, 7, 12, \ldots$ is 62? 2

14. Find the sum of the first 20 terms of the AP $6, \frac{15}{2}, 9, \frac{21}{2}, \ldots$ 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an AP, the 4th term is 3 and the 9th term is -12. Find the AP and its 14th term. 3

16. Find the sum of the first 12 terms of the AP whose nth term is $a_n = 4n + 7$. 3

17. The sum of the first 15 terms of an AP is 660 and the first term is 2. Find d and the 15th term. 3

OR

In an AP, the 3rd term is 18 and the 6th term is 33. Find the AP and its 20th term. 3

18. Find the sum of the first 25 terms of the AP whose nth term is $a_n = 2n + 9$. 3

OR

The sum of the first 20 terms of an AP is 600 and the first term is 11. Find d and the 20th term. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. How many terms of the AP $-10, -6, -2, 2, \ldots$ must be taken so that the sum is 144? 5

OR

A ladder has 7 rungs whose lengths decrease uniformly from 50 cm at the bottom to 20 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

20. How many terms of the AP $36, 33, 30, 27, \ldots$ must be taken so that the sum is 231? 5

OR

A ladder has 9 rungs whose lengths decrease uniformly from 80 cm at the bottom to 40 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

END OF QUESTION PAPER — Total: 40 Marks

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